PRACTICAL 1
Installing RHEL and Studying the Linux Graphical Environment

Aim

To install Red Hat Enterprise Linux in VirtualBox/VMware and study the graphical environment, login manager, desktop environment, directory structure and file systems.

Requirements

- VirtualBox or VMware
- RHEL ISO
- 4 GB RAM or more
- 30 GB virtual disk

A. Creating the Virtual Machine

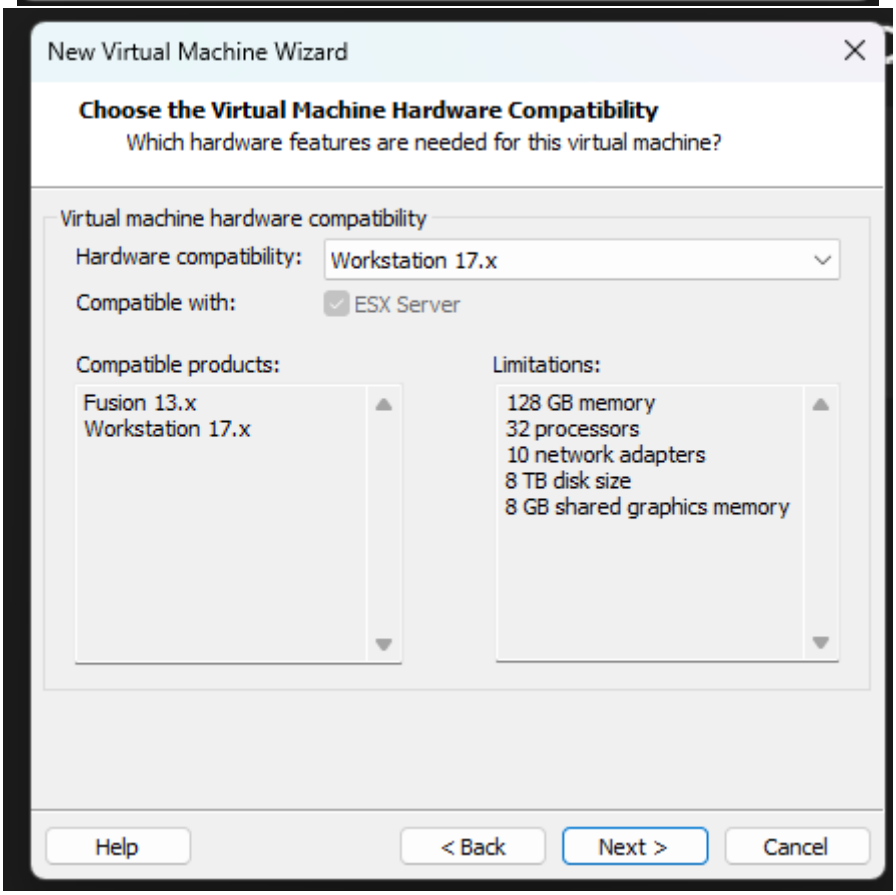
Procedure

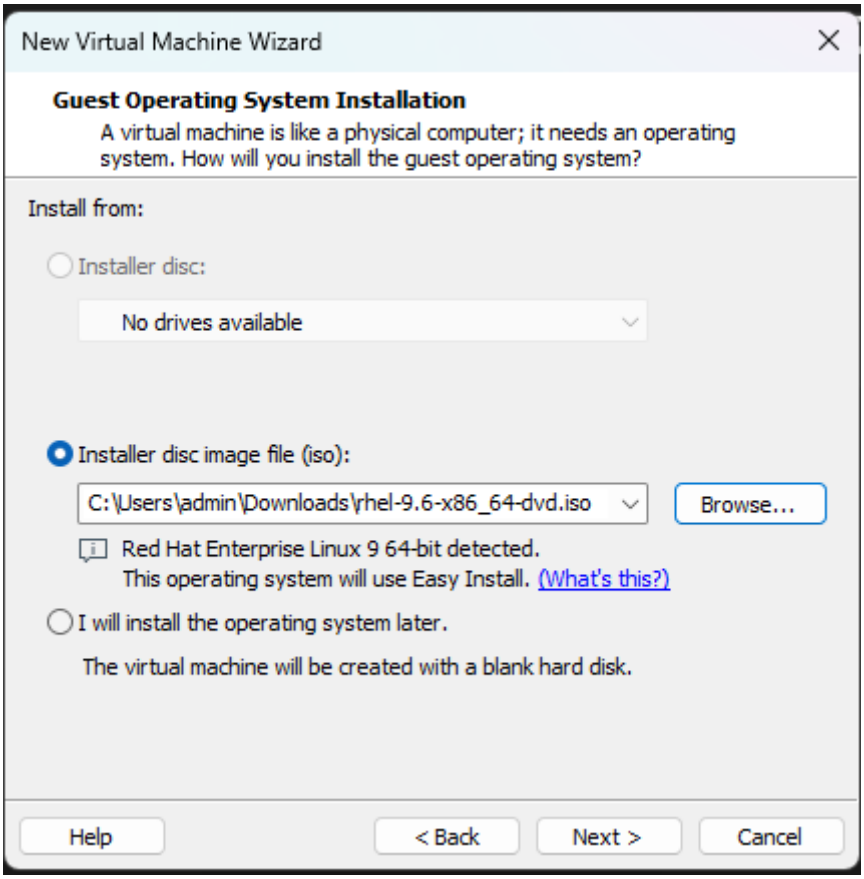
1. Open VirtualBox/VMware.
2. Create a new virtual machine.
3. Select Linux / Red Hat Enterprise Linux 64-bit.
4. Allocate RAM.
5. Create a virtual hard disk.
6. Attach the RHEL ISO.
7. Start the VM.
8. Select **Install Red Hat Enterprise Linux**.
9. Select language and keyboard.
10. Configure:

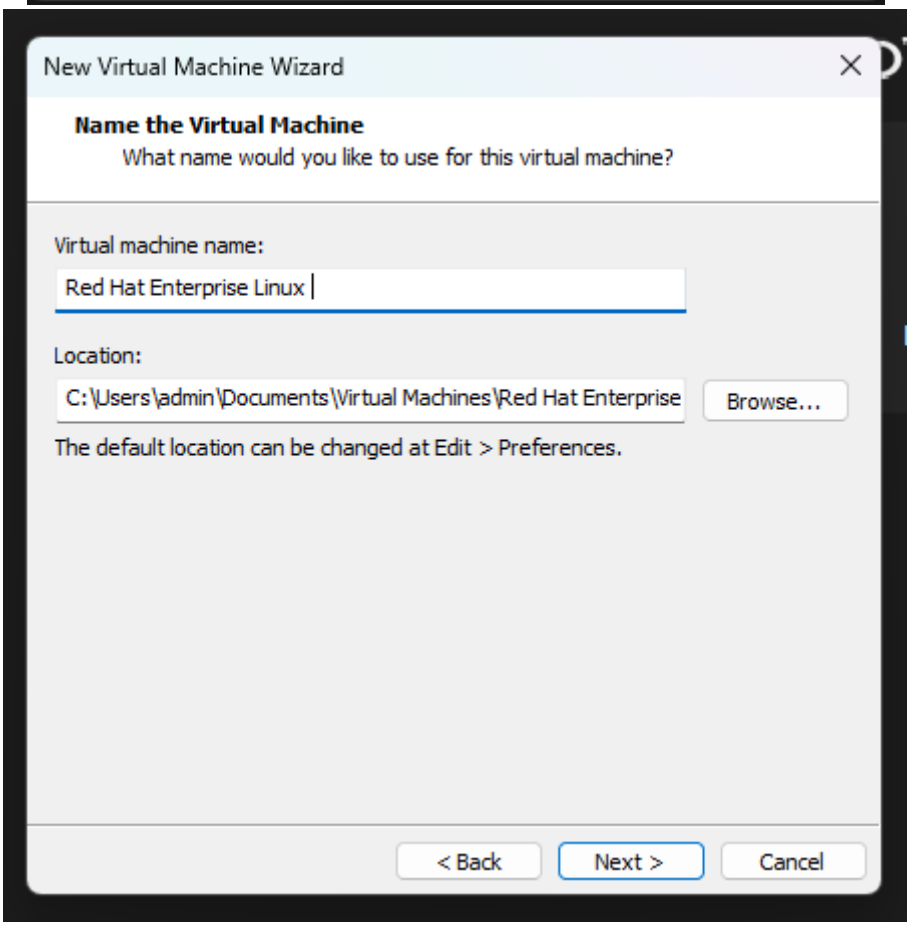
- Installation destination
- Network
- Time zone
- User account
- Root password

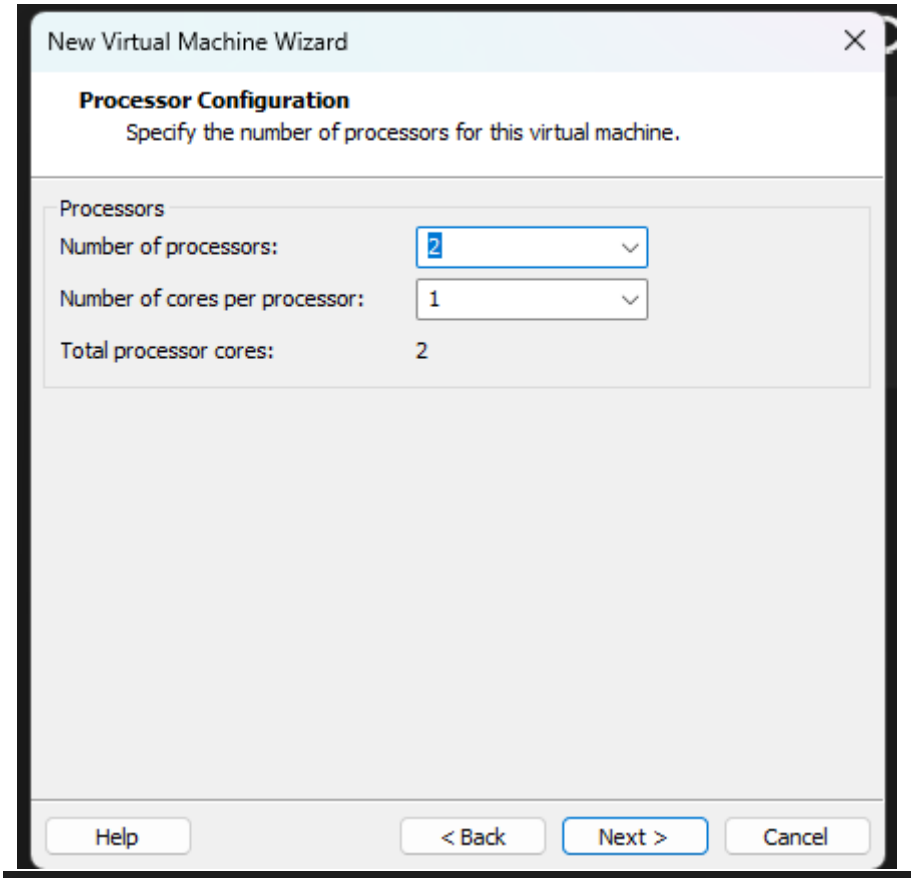
11. Begin installation.
12. Reboot after installation.
13. Log in.

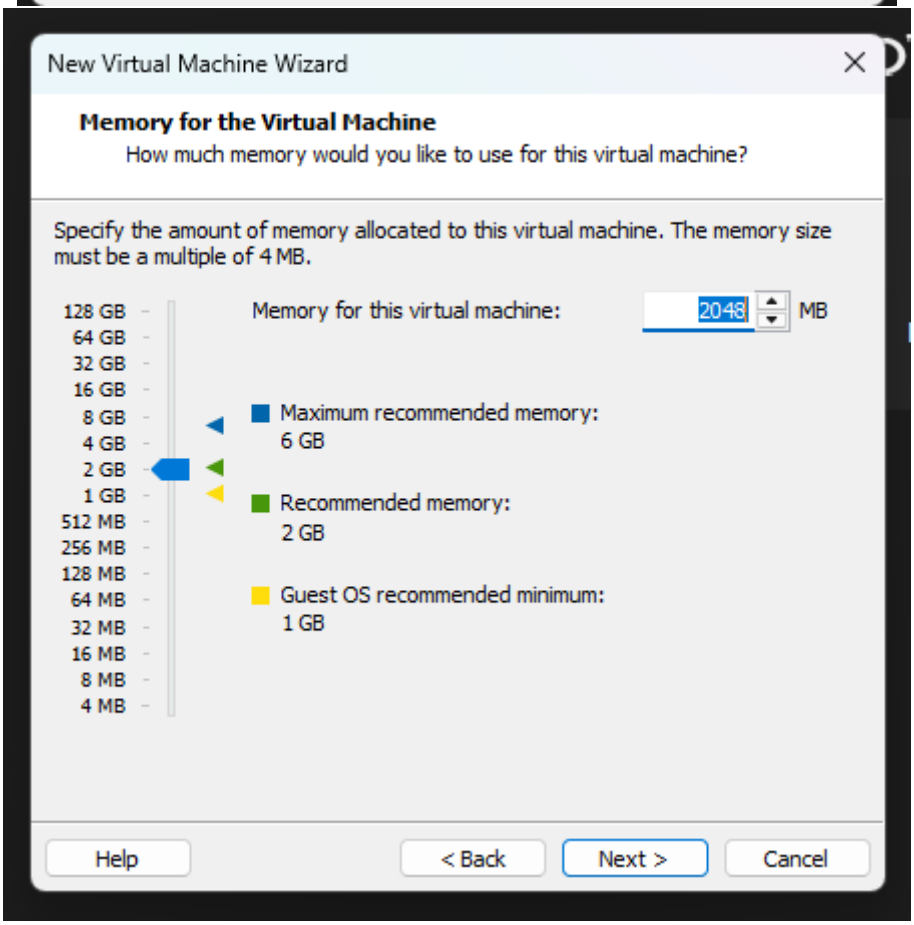

New Virtual Machine Wizard
vmware
WORKSTATION
PRO™
17
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation 17.x virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.
Help
< Back
Next >
Cancel

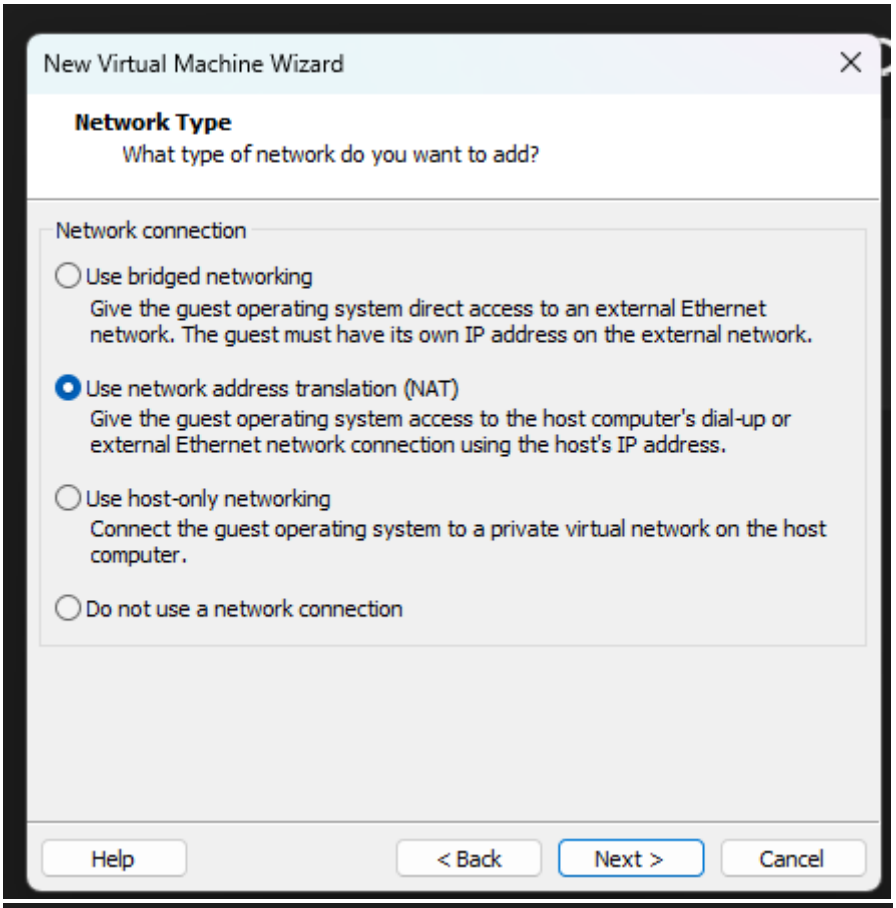

New Virtual Machine Wizard
Choose the Virtual Machine Hardware Compatibility
Which hardware features are needed for this virtual machine?
Virtual machine hardware compatibility
Hardware compatibility:
Workstation 17.x
Compatible with:
ESX Server
Compatible products:
Fusion 13.x
Workstation 17.x
Limitations:
128 GB memory
32 processors
10 network adapters
8 TB disk size
8 GB shared graphics memory
Help
< Back
Next >
Cancel

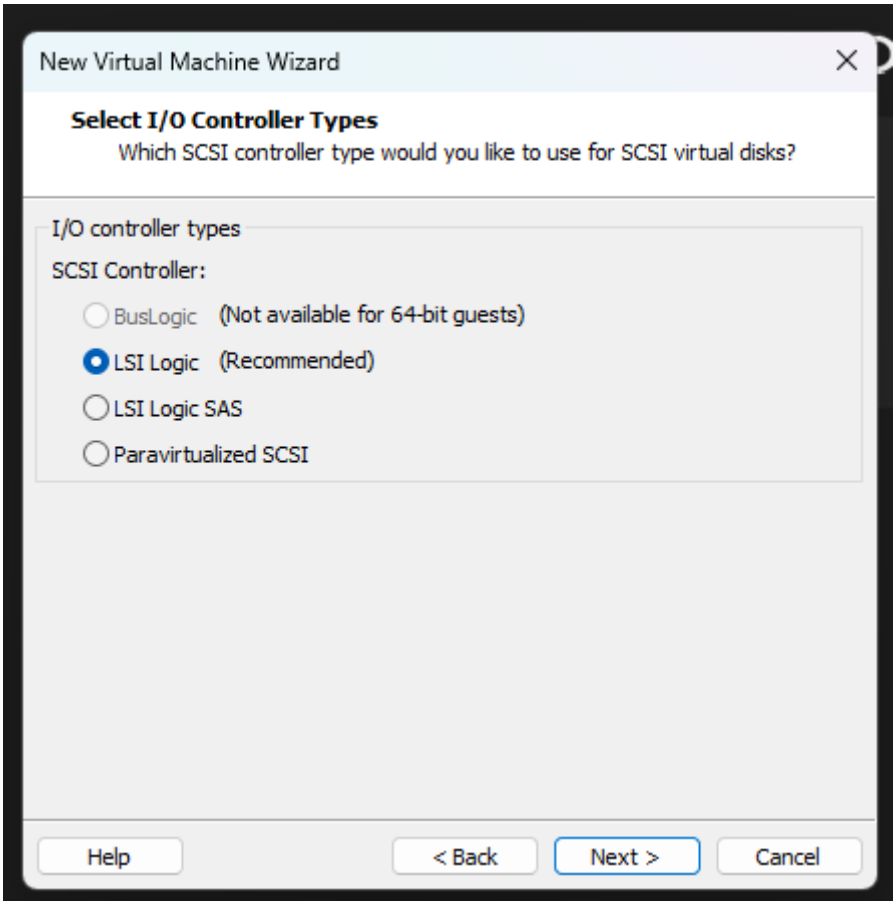

New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\Users\admin\Downloads\rhel-9.6-x86_64-dvd.iso
Browse...
Red Hat Enterprise Linux 9 64-bit detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel

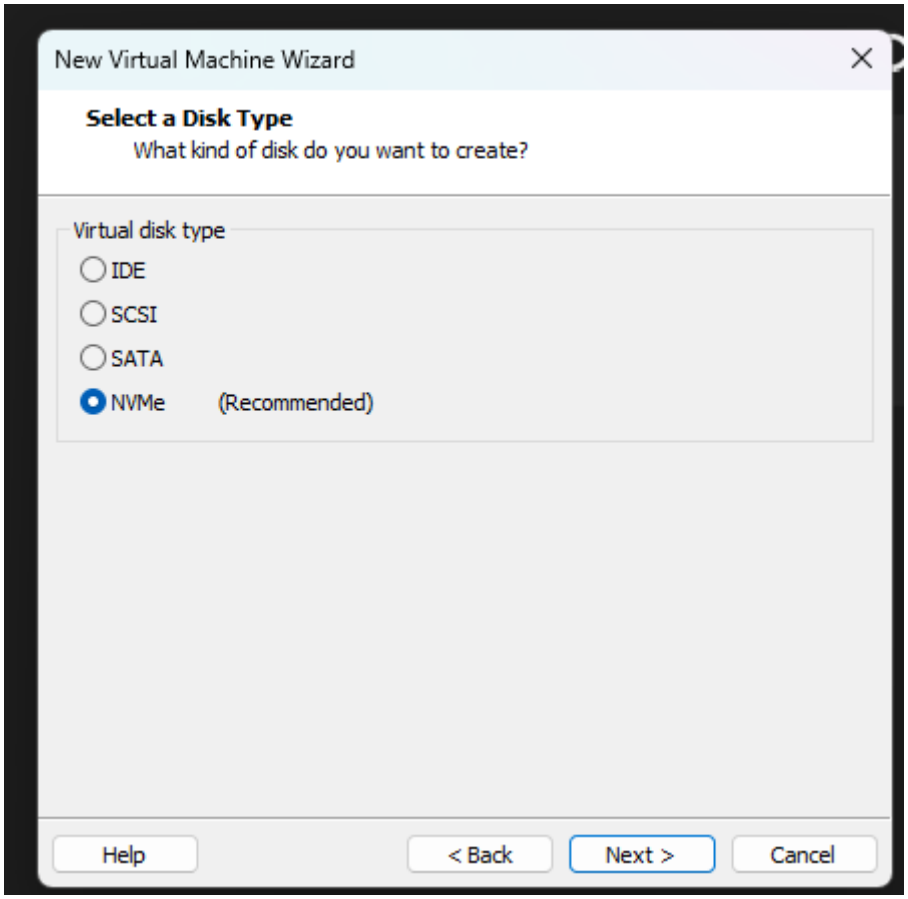

New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Red Hat Enterprise Linux
Location:
C:\Users\admin\Documents\Virtual Machines\Red Hat Enterprise
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel

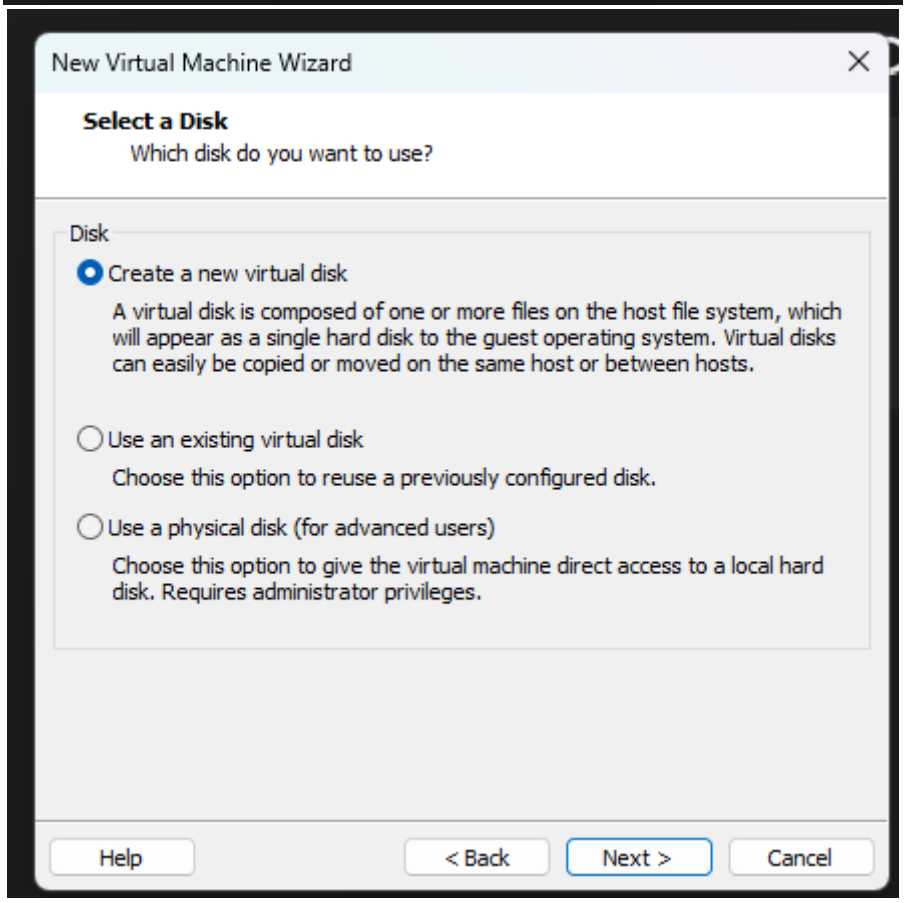

New Virtual Machine Wizard
Processor Configuration
Specify the number of processors for this virtual machine.
Processors
Number of processors:
2
Number of cores per processor:
1
Total processor cores:
2
Help
< Back
Next >
Cancel

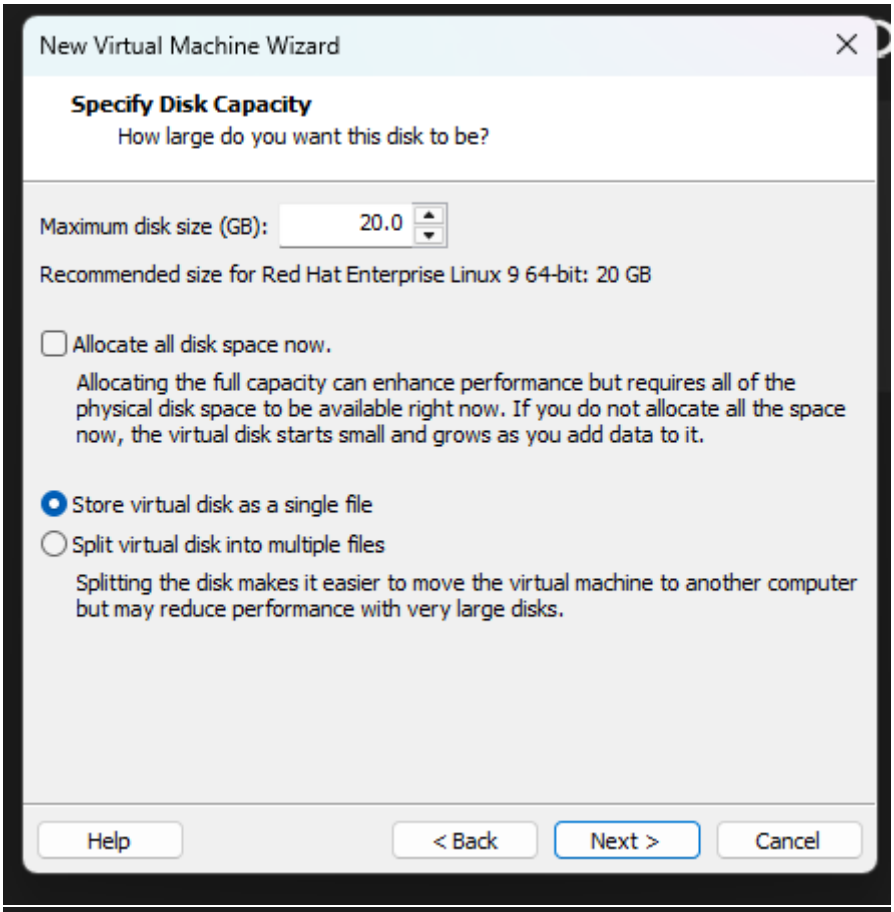

New Virtual Machine Wizard
Memory for the Virtual Machine
How much memory would you like to use for this virtual machine?
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Memory for this virtual machine:
2048
MB
Maximum recommended memory:
6 GB
Recommended memory:
2 GB
Guest OS recommended minimum:
1 GB
Help
< Back
Next >
Cancel

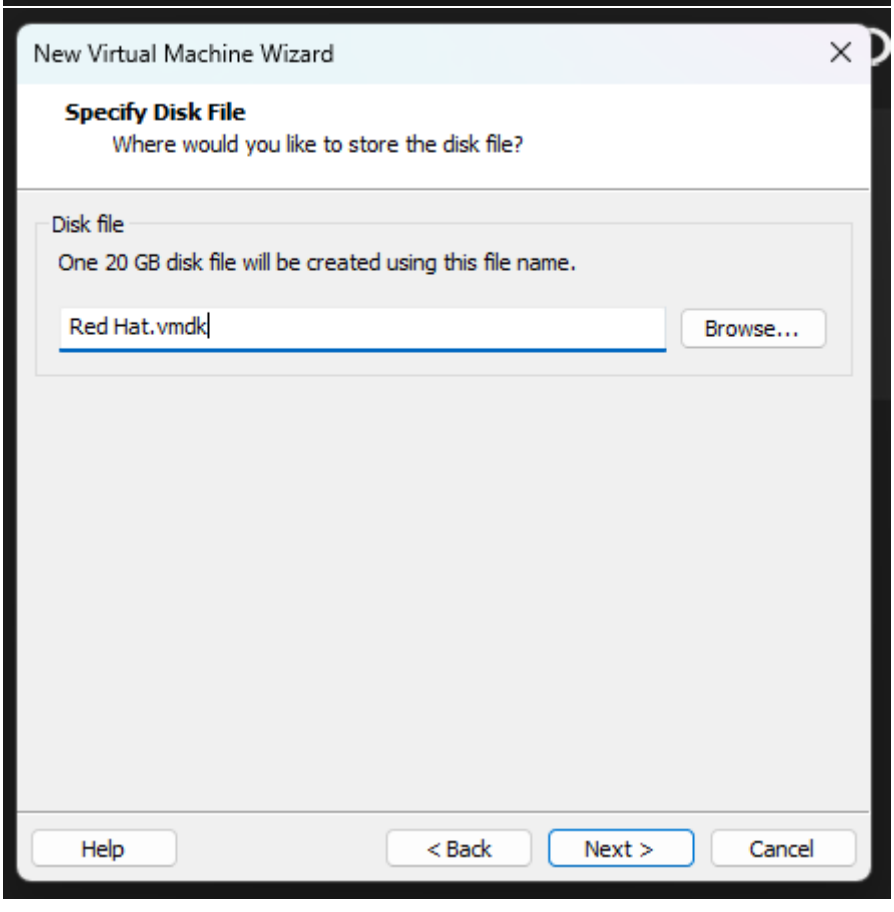

New Virtual Machine Wizard
Network Type
What type of network do you want to add?
Network connection
Use bridged networking
Give the guest operating system direct access to an external Ethernet network. The guest must have its own IP address on the external network.
Use network address translation (NAT)
Give the guest operating system access to the host computer's dial-up or external Ethernet network connection using the host's IP address.
Use host-only networking
Connect the guest operating system to a private virtual network on the host computer.
Do not use a network connection
Help
< Back
Next >
Cancel

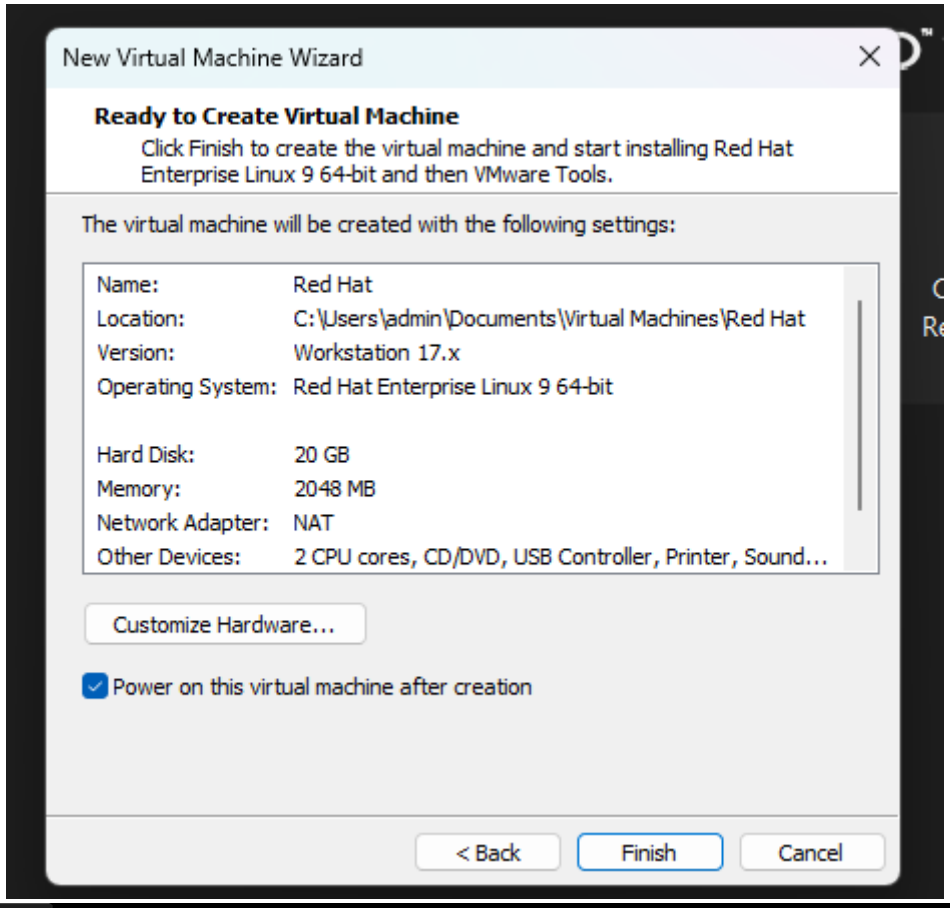

New Virtual Machine Wizard
Select I/O Controller Types
Which SCSI controller type would you like to use for SCSI virtual disks?
I/O controller types
SCSI Controller:
BusLogic (Not available for 64-bit guests)
LSI Logic (Recommended)
LSI Logic SAS
Paravirtualized SCSI
Help
< Back
Next >
Cancel

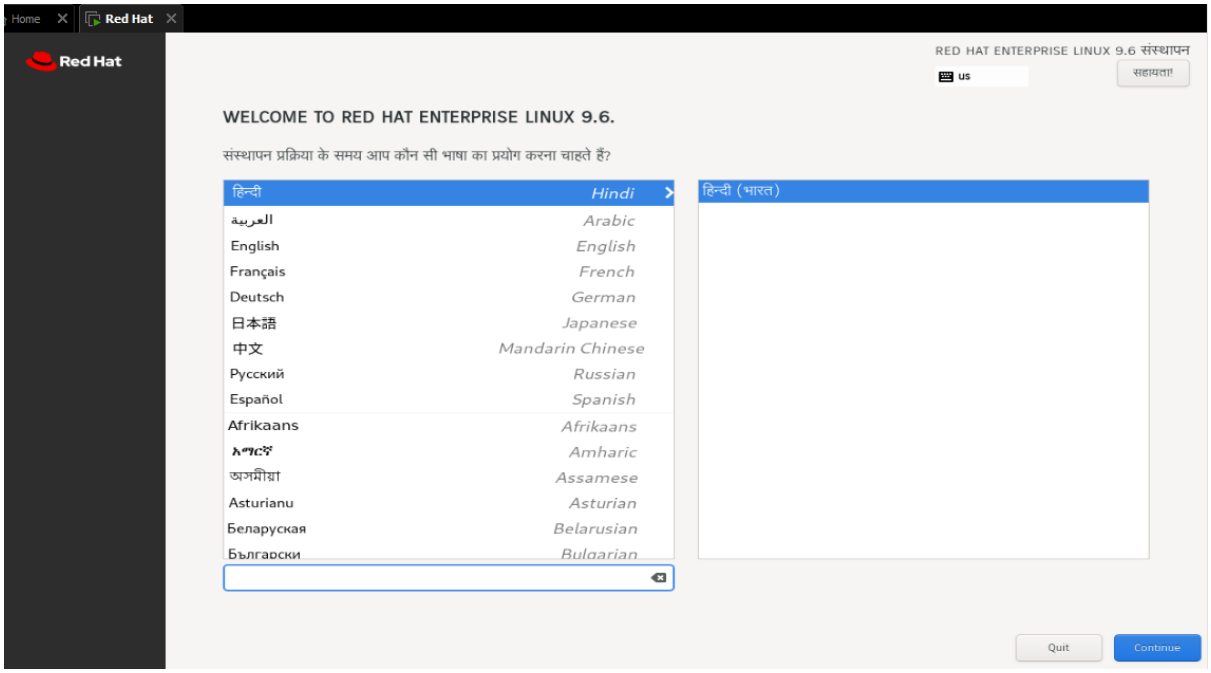

New Virtual Machine Wizard
Select a Disk Type
What kind of disk do you want to create?
Virtual disk type
IDE
SCSI
SATA
NVMe (Recommended)
Help
< Back
Next >
Cancel

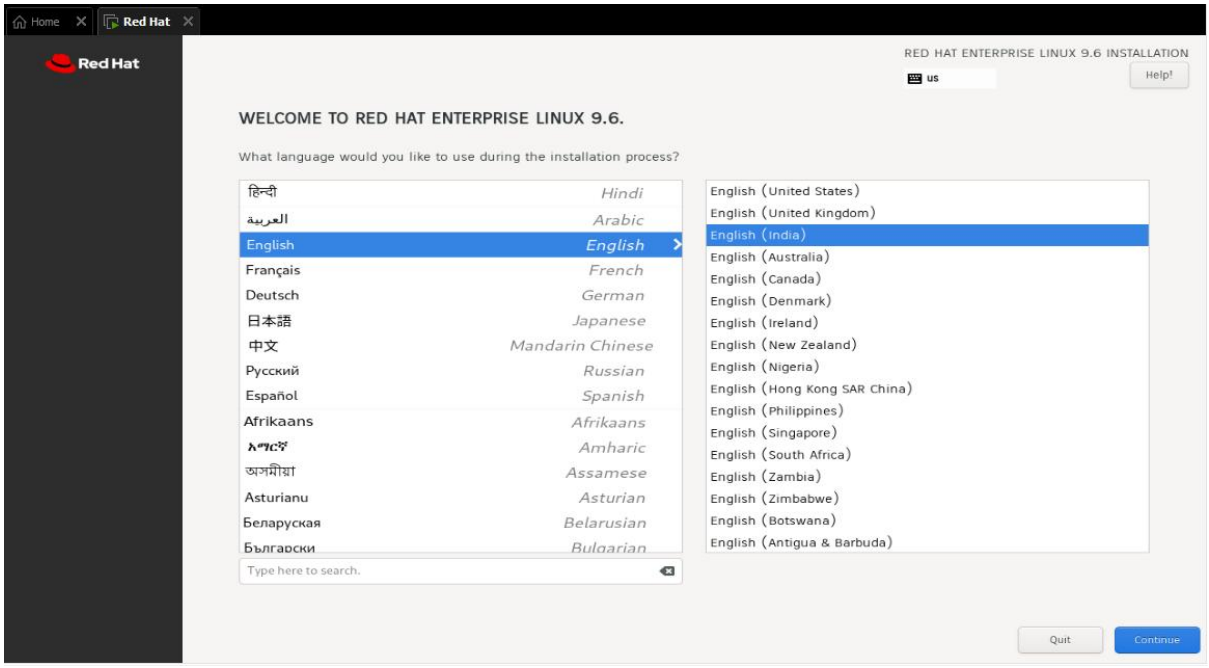

New Virtual Machine Wizard
Select a Disk
Which disk do you want to use?
Disk
Create a new virtual disk
A virtual disk is composed of one or more files on the host file system, which will appear as a single hard disk to the guest operating system. Virtual disks can easily be copied or moved on the same host or between hosts.
Use an existing virtual disk
Choose this option to reuse a previously configured disk.
Use a physical disk (for advanced users)
Choose this option to give the virtual machine direct access to a local hard disk. Requires administrator privileges.
Help
< Back
Next >
Cancel

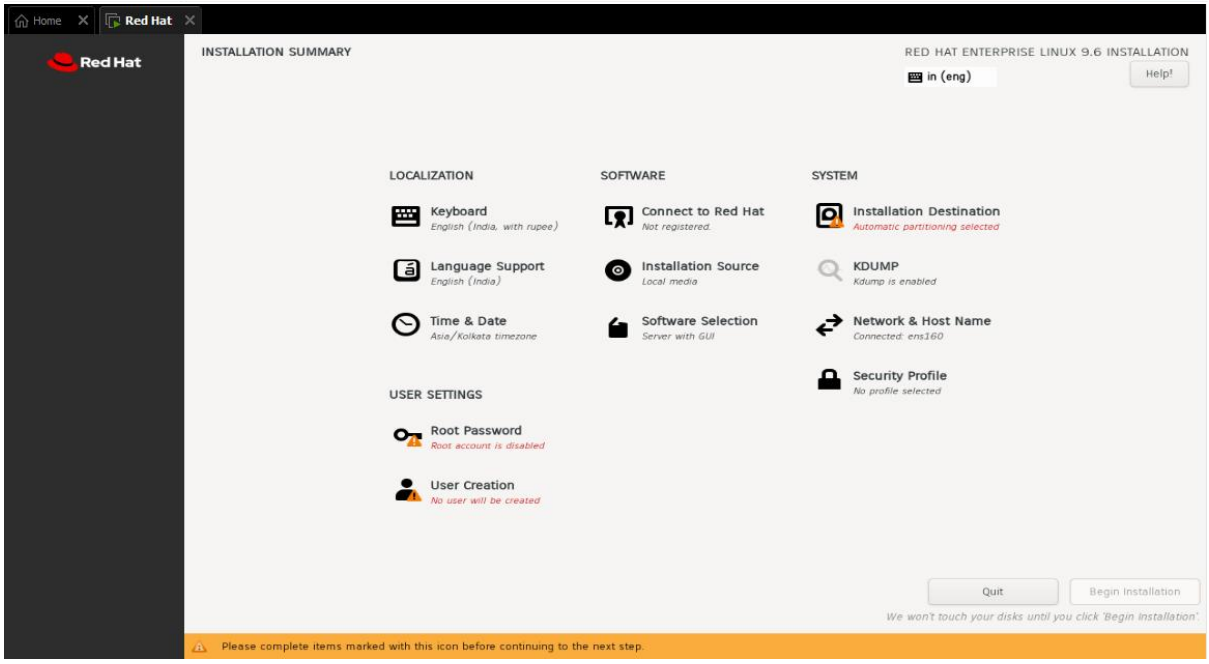

New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
Maximum disk size (GB):
20.0
Recommended size for Red Hat Enterprise Linux 9 64-bit: 20 GB
Allocate all disk space now.
Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel

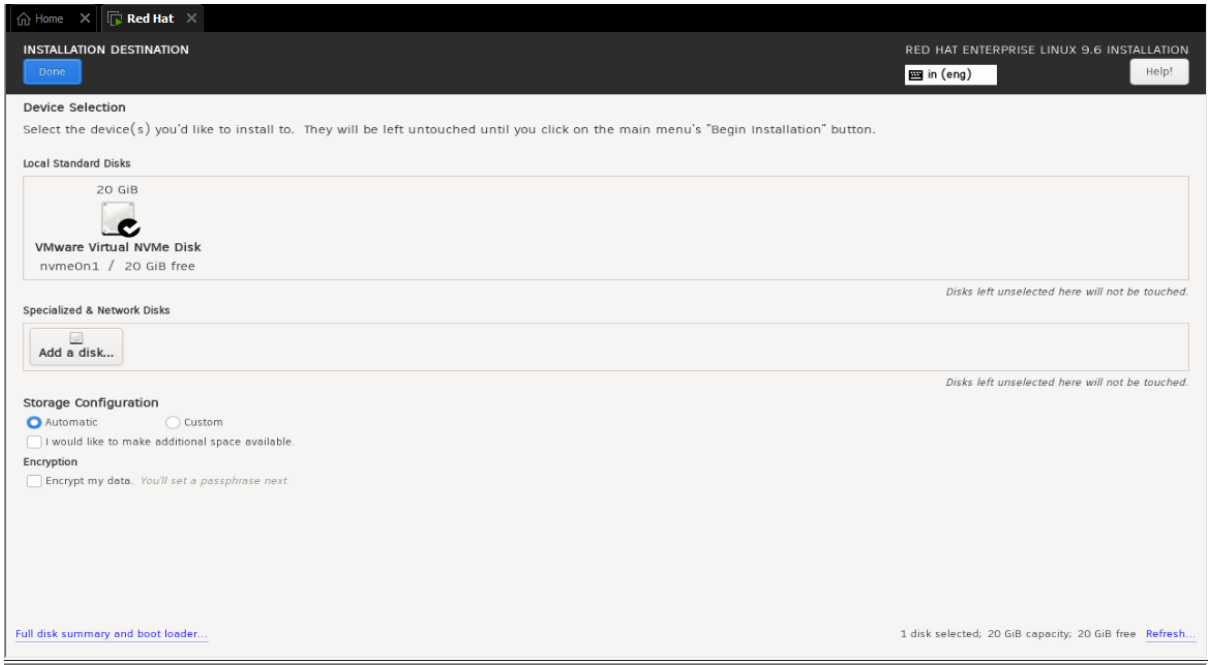

New Virtual Machine Wizard
Specify Disk File
Where would you like to store the disk file?
Disk file
One 20 GB disk file will be created using this file name.
Red Hat.vmdk
Browse...
Help
< Back
Next >
Cancel

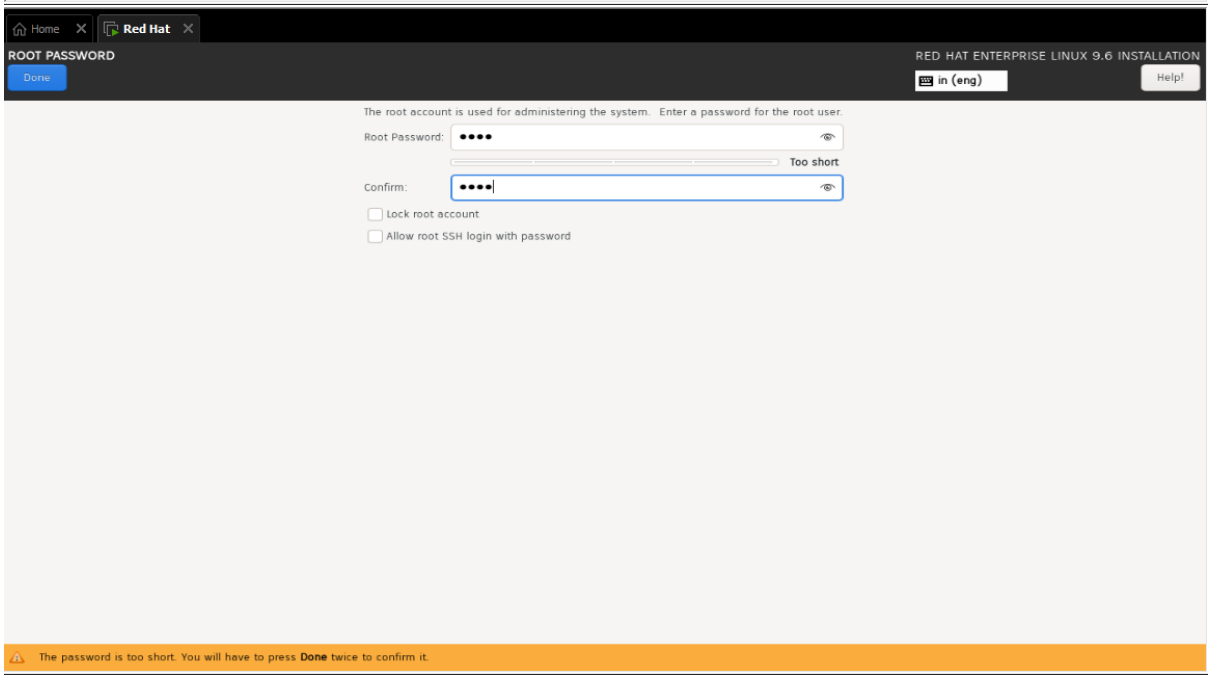

New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Red Hat Enterprise Linux 9 64-bit and then VMware Tools.
The virtual machine will be created with the following settings:
Name: Red Hat
Location: C:\Users\admin\Documents\Virtual Machines\Red Hat
Version: Workstation 17.x
Operating System: Red Hat Enterprise Linux 9 64-bit
Hard Disk: 20 GB
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound...
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel

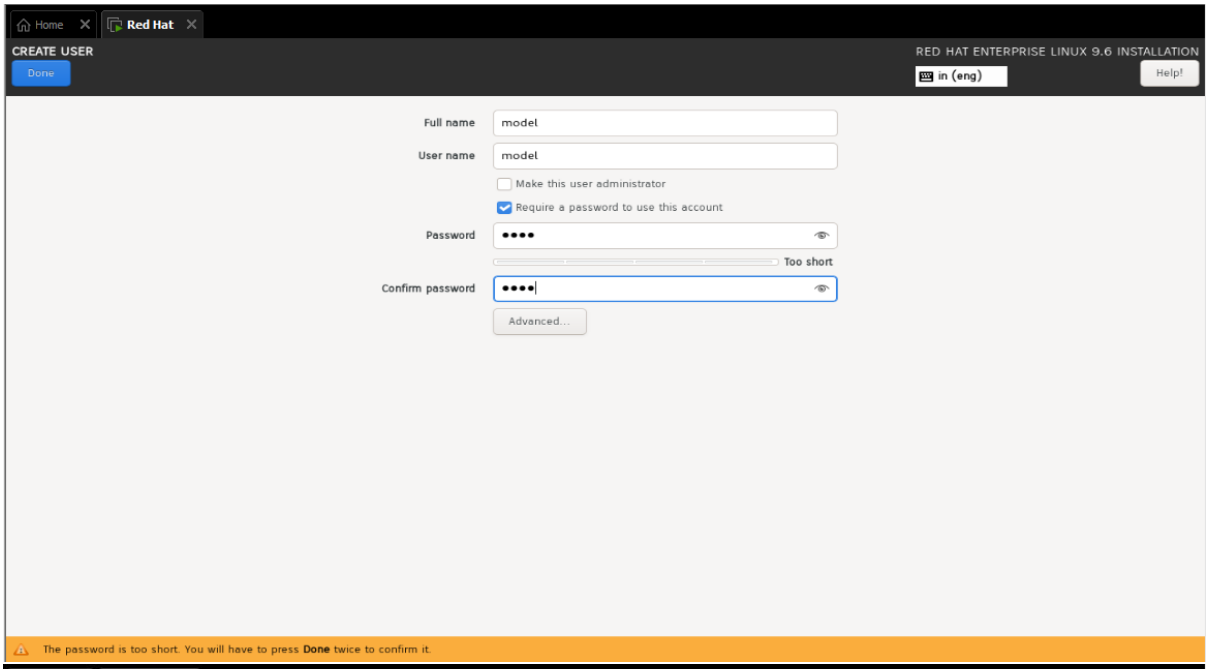

Home
Red Hat
Red Hat
RED HAT ENTERPRISE LINUX 9.6 संस्थापन
us
सहायता!
WELCOME TO RED HAT ENTERPRISE LINUX 9.6.
संस्थापन प्रक्रिया के समय आप कौन सी भाषा का प्रयोग करना चाहते हैं?
हिन्दी Hindi
العربية Arabic
English English
Français French
Deutsch German
日本語 Japanese
中文 Mandarin Chinese
Русский Russian
Español Spanish
Afrikaans Afrikaans
አማርኛ Amharic
অসমীয়া Assamese
Asturianu Asturian
Беларуская Belarusian
Български Bulgarian
हिन्दी (भारत)
Quit
Continue

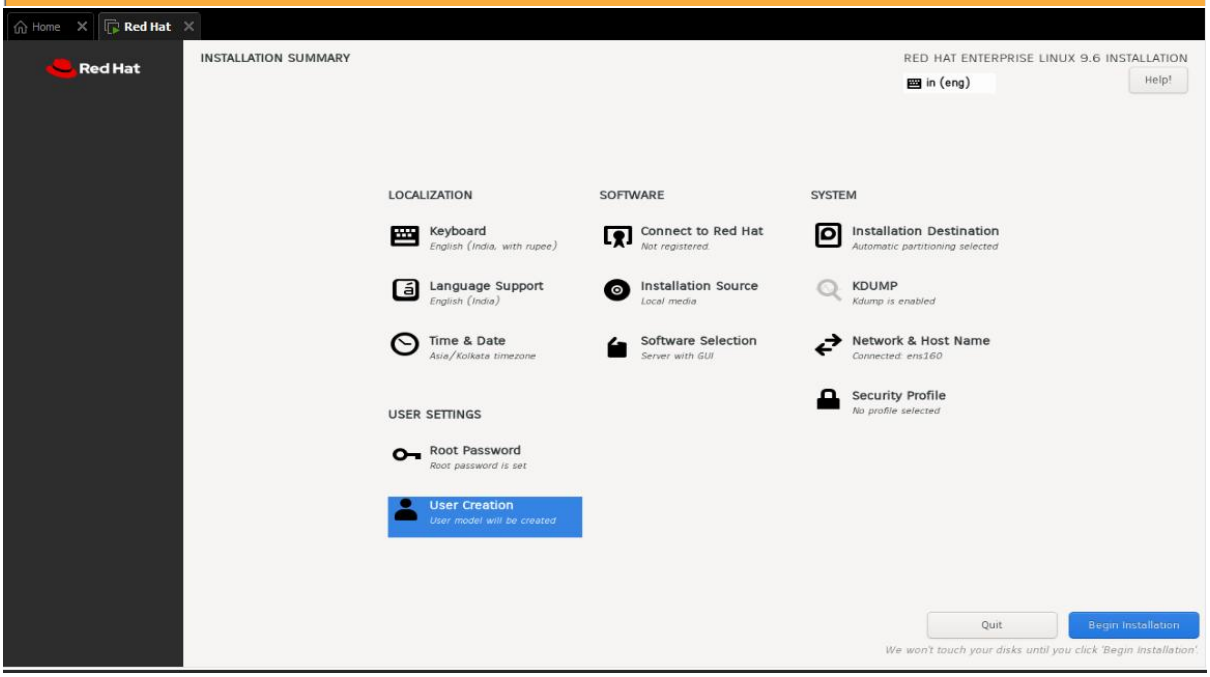

Home
Red Hat
Red Hat
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
us
Help!
WELCOME TO RED HAT ENTERPRISE LINUX 9.6.
What language would you like to use during the installation process?
हिन्दी Hindi
العربية Arabic
English English
Français French
Deutsch German
日本語 Japanese
中文 Mandarin Chinese
Русский Russian
Español Spanish
Afrikaans Afrikaans
አማርኛ Amharic
অসমীয়া Assamese
Asturianu Asturian
Беларуская Belarusian
Български Bulgarian
Type here to search.
English (United States)
English (United Kingdom)
English (India)
English (Australia)
English (Canada)
English (Denmark)
English (Ireland)
English (New Zealand)
English (Nigeria)
English (Hong Kong SAR China)
English (Philippines)
English (Singapore)
English (South Africa)
English (Zambia)
English (Zimbabwe)
English (Botswana)
English (Antigua & Barbuda)
Quit
Continue

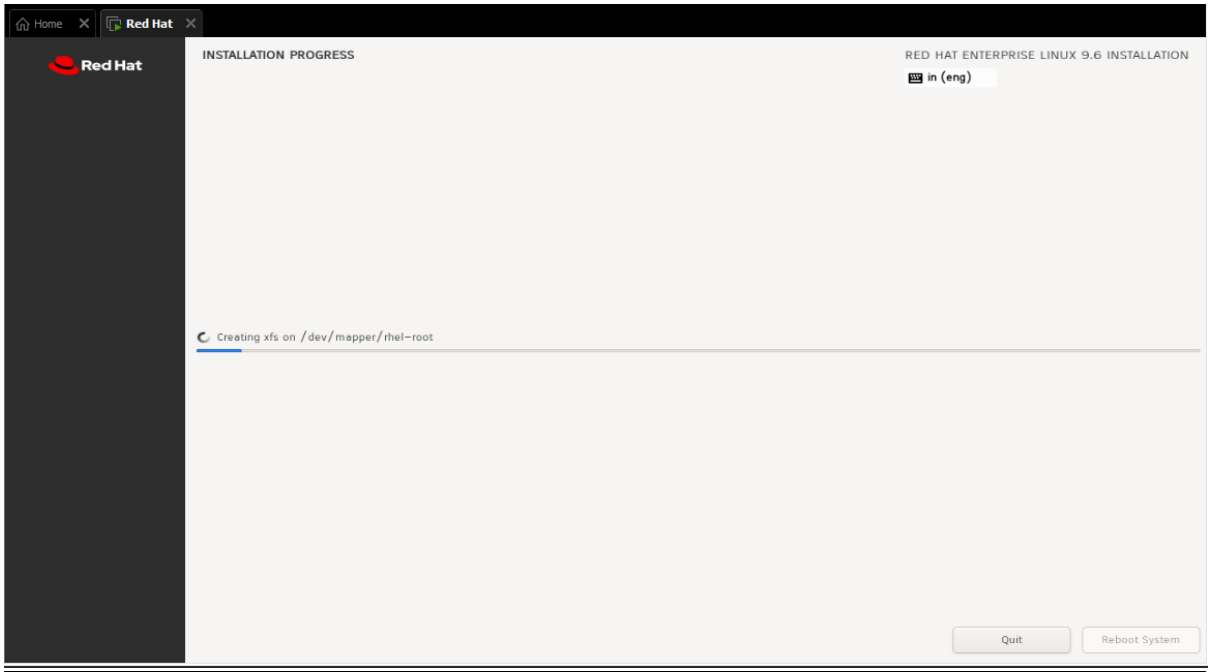

Home
Red Hat
Red Hat
INSTALLATION SUMMARY
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
LOCALIZATION
Keyboard
English (India, with rupee)
Language Support
English (India)
Time & Date
Asia/Kolkata timezone
SOFTWARE
Connect to Red Hat
Not registered.
Installation Source
Local media
Software Selection
Server with GUI
SYSTEM
Installation Destination
Automatic partitioning selected
KDUMP
Kdump is enabled
Network & Host Name
Connected: ens160
Security Profile
No profile selected
USER SETTINGS
Root Password
Root account is disabled
User Creation
No user will be created
Quit
Begin Installation
We won't touch your disks until you click 'Begin Installation'.
Please complete items marked with this icon before continuing to the next step.

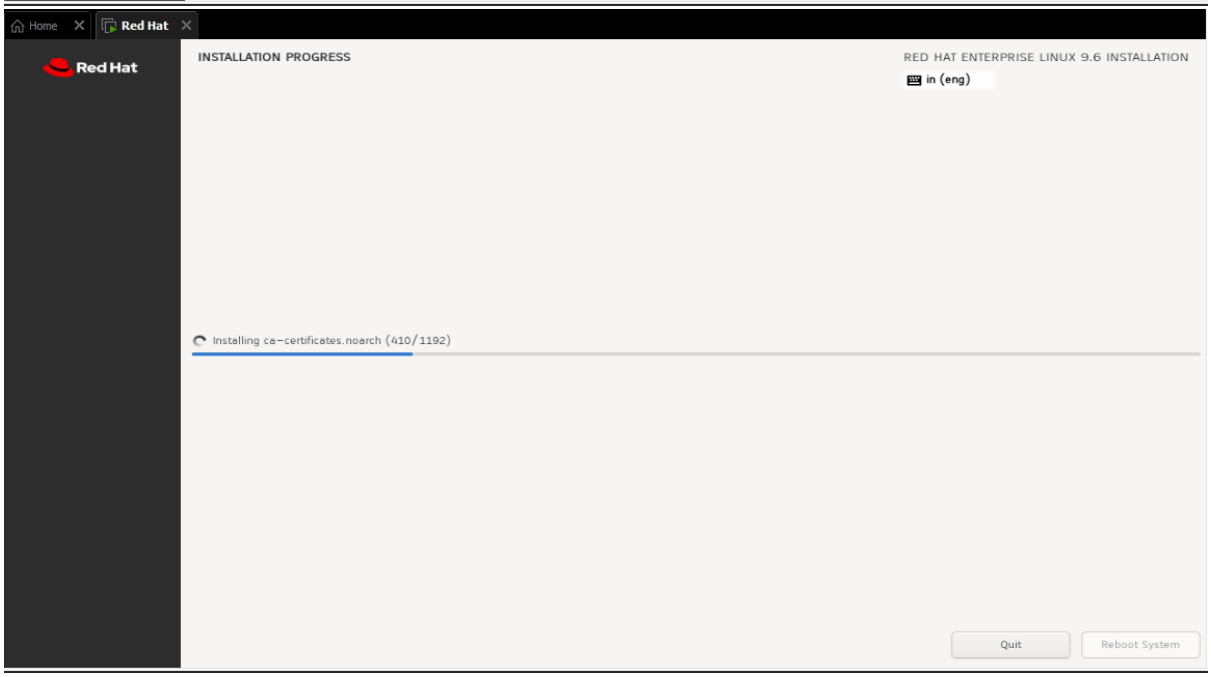

Home
Red Hat
INSTALLATION DESTINATION
Done
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
Device Selection
Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.
Local Standard Disks
20 GiB
VMware Virtual NVMe Disk
nvme0n1 / 20 GiB free
Disks left unselected here will not be touched.
Specialized & Network Disks
Add a disk...
Disks left unselected here will not be touched.
Storage Configuration
Automatic
Custom
I would like to make additional space available.
Encryption
Encrypt my data. You'll set a passphrase next.
Full disk summary and boot loader...
1 disk selected; 20 GiB capacity; 20 GiB free
Refresh...


Home
Red Hat
ROOT PASSWORD
Done
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
The root account is used for administering the system. Enter a password for the root user.
Root Password:
Too short
Confirm:
Lock root account
Allow root SSH login with password
The password is too short. You will have to press Done twice to confirm it.

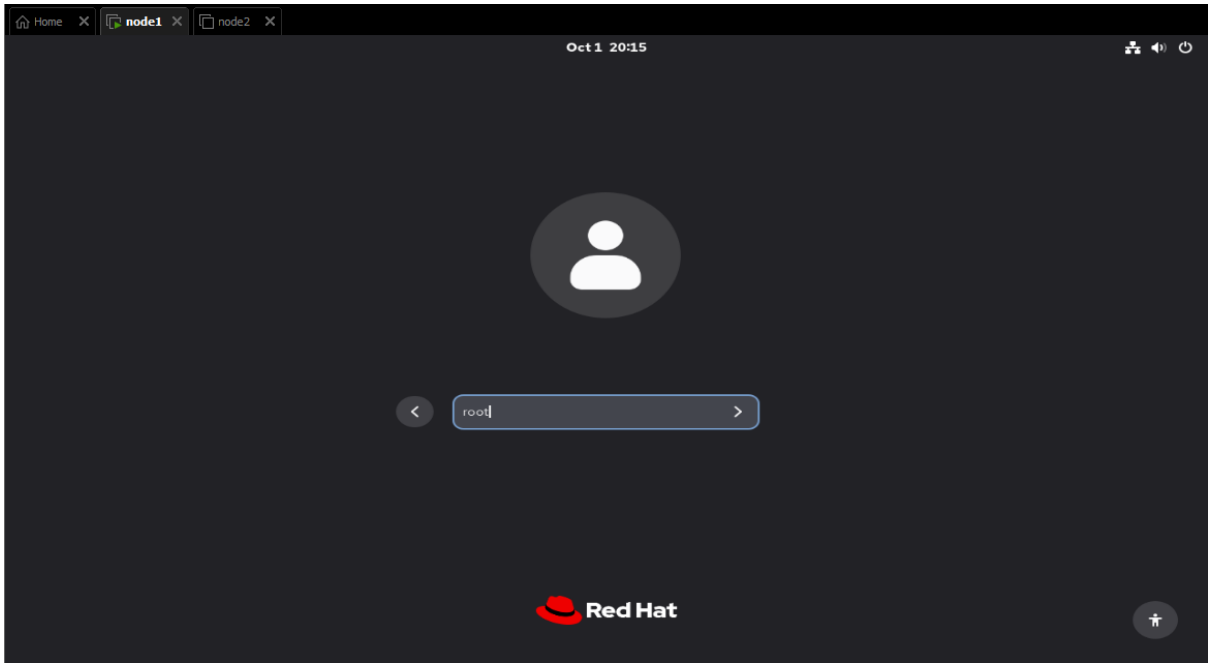

Home
Red Hat
CREATE USER
Done
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
Full name
model
User name
model
Make this user administrator
Require a password to use this account
Password
Too short
Confirm password
Advanced...
The password is too short. You will have to press Done twice to confirm it.

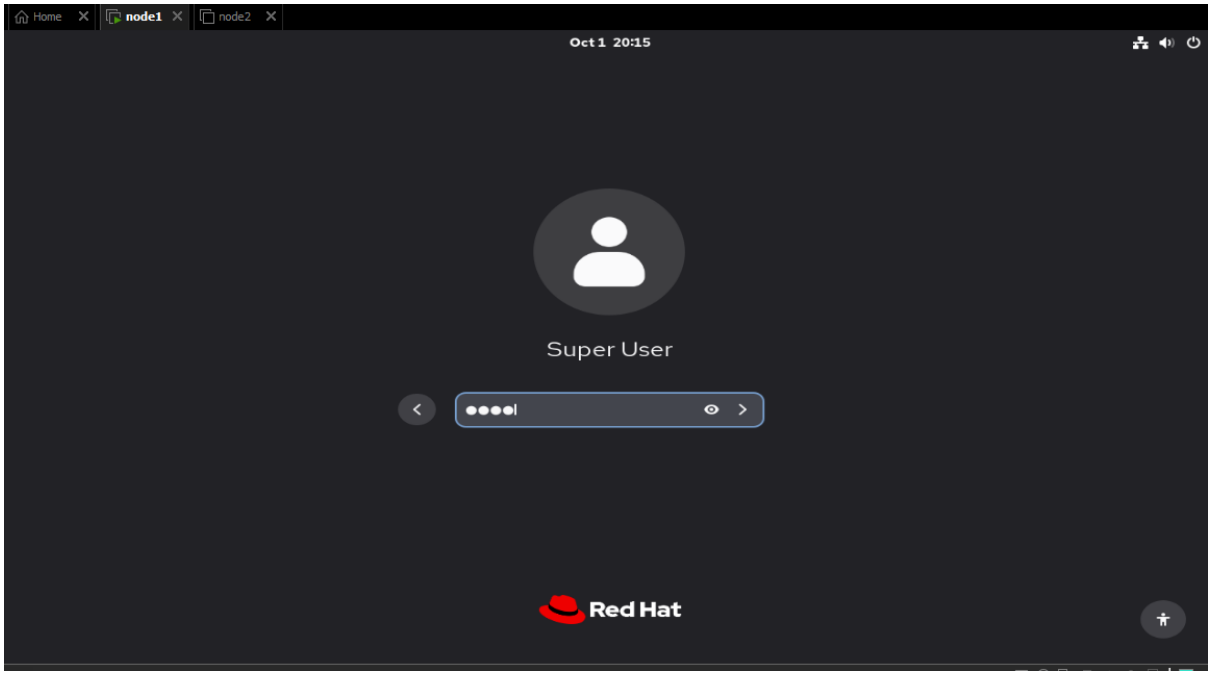

Home
Red Hat
Red Hat
INSTALLATION SUMMARY
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
LOCALIZATION
Keyboard
English (India, with rupee)
Language Support
English (India)
Time & Date
Asia/Kolkata timezone
SOFTWARE
Connect to Red Hat
Not registered.
Installation Source
Local media
Software Selection
Server with GUI
SYSTEM
Installation Destination
Automatic partitioning selected
KDUMP
Kdump is enabled
Network & Host Name
Connected: ens160
Security Profile
No profile selected
USER SETTINGS
Root Password
Root password is set
User Creation
User model will be created
Quit
Begin Installation
We won't touch your disks until you click 'Begin Installation'.


Home
Red Hat
Red Hat
INSTALLATION PROGRESS
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Creating xfs on /dev/mapper/rhel-root
Quit
Reboot System


Home
Red Hat
Red Hat
INSTALLATION PROGRESS
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Installing ca-certificates.noarch (410/1192)
Quit
Reboot System


Home
Red Hat
Red Hat
INSTALLATION PROGRESS
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Complete!
Red Hat Enterprise Linux is now successfully installed and ready for you to use!
Go ahead and reboot your system to start using it!
Reboot System
Use of this product is subject to the license agreement found at /usr/share/redhat-release/EULA


Home
node1
node2
Oct 1 20:15
root
Red Hat


Home
node1
node2
Oct 1 20:15
Super User
Red Hat



Successfully logged-in RHEL desktop.

B. Linux Graphical Environment

1. Display/Login Manager

Check the display manager:

systemctl status display-manager

Expected Output

● gdm.service - GNOME Display Manager

Loaded: loaded

Active: active (running)

Run:

systemctl status display-manager

Print this terminal output.



---

2. Desktop Environment

RHEL Workstation commonly uses GNOME.

Execute:
echo $XDG_CURRENT_DESKTOP
echo $DESKTOP_SESSION
Expected Output
GNOME
gnome
Check operating system:
cat /etc/os-release
Expected Output
NAME="Red Hat Enterprise Linux"
VERSION="10..."
ID="rhel"



C. Linux Directory Structure
Execute:
ls /
Typical directories include:
bin
boot
dev
etc
home
lib
lib64
media
mnt
opt
proc
root
run

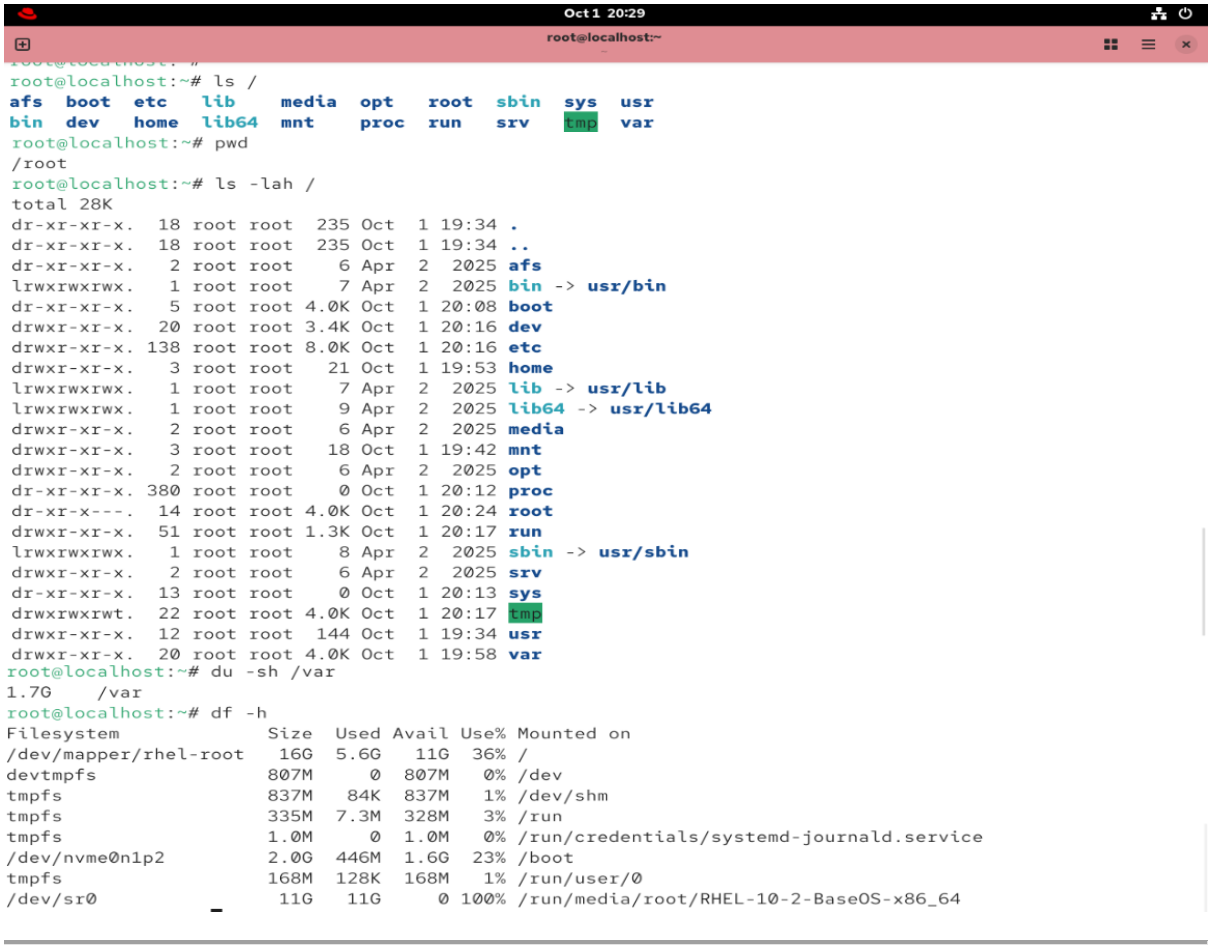
sbin
srv
sys
tmp
usr
var

Important Directories

Directory	Purpose
/	Root directory
/boot	Boot-related files
/etc	System configuration
/home	User home directories
/root	Root user's home
/tmp	Temporary files
/usr	User applications and system resources
/var	Variable data/logs
/dev	Device files
/proc	Process/kernel information
/sys	Kernel/device information
/mnt	Temporary mount point
/opt	Optional software

Commands

pwd
ls -lah /
du -sh /var
df -h


Oct 1 20:29
root@localhost:~
root@localhost:~# ls /
afs boot etc lib media opt root sbin sys usr
bin dev home lib64 mnt proc run srv tmp var
root@localhost:~# pwd
/root
root@localhost:~# ls -lah /
total 28K
dr-xr-xr-x. 18 root root 235 Oct 1 19:34 .
dr-xr-xr-x. 18 root root 235 Oct 1 19:34 ..
dr-xr-xr-x. 2 root root 6 Apr 2 2025 afs
lrwxrwxrwx. 1 root root 7 Apr 2 2025 bin -> usr/bin
dr-xr-xr-x. 5 root root 4.0K Oct 1 20:08 boot
drwxr-xr-x. 20 root root 3.4K Oct 1 20:16 dev
drwxr-xr-x. 138 root root 8.0K Oct 1 20:16 etc
drwxr-xr-x. 3 root root 21 Oct 1 19:53 home
lrwxrwxrwx. 1 root root 7 Apr 2 2025 lib -> usr/lib
lrwxrwxrwx. 1 root root 9 Apr 2 2025 lib64 -> usr/lib64
drwxr-xr-x. 2 root root 6 Apr 2 2025 media
drwxr-xr-x. 3 root root 18 Oct 1 19:42 mnt
drwxr-xr-x. 2 root root 6 Apr 2 2025 opt
dr-xr-xr-x. 380 root root 0 Oct 1 20:12 proc
dr-xr-x---. 14 root root 4.0K Oct 1 20:24 root
drwxr-xr-x. 51 root root 1.3K Oct 1 20:17 run
lrwxrwxrwx. 1 root root 8 Apr 2 2025 sbin -> usr/sbin
drwxr-xr-x. 2 root root 6 Apr 2 2025 srv
dr-xr-xr-x. 13 root root 0 Oct 1 20:13 sys
drwxrwxrwt. 22 root root 4.0K Oct 1 20:17 tmp
drwxr-xr-x. 12 root root 144 Oct 1 19:34 usr
drwxr-xr-x. 20 root root 4.0K Oct 1 19:58 var
root@localhost:~# du -sh /var
1.7G /var
root@localhost:~# df -h
Filesystem Size Used Avail Use% Mounted on
/dev/mapper/rhel-root 16G 5.6G 11G 36% /
devtmpfs 807M 0 807M 0% /dev
tmpfs 837M 84K 837M 1% /dev/shm
tmpfs 335M 7.3M 328M 3% /run
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2 2.0G 446M 1.6G 23% /boot
tmpfs 168M 128K 168M 1% /run/user/0
/dev/sr0 11G 11G 0 100% /run/media/root/RHEL-10-2-BaseOS-x86_64